LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

Lesson 8: Coordinates and Random: Things That Fall

Module 3 | 45 minutes | Coordinates and randomness | Grades 3–5 · ages 8–10

Objective: Use x and y coordinates and “pick random” to place sprites, and make an object fall and restart from a random place at the top.

Success criteria

- I can place a sprite using x and y.
- I can make a sprite fall and restart at the top.
- I can use pick random.

Key vocabulary

- **coordinates**: Numbers (x, y) that say exactly where something is.
- **x**: Left–right position.
- **y**: Up–down position.
- **random**: Chosen by chance, not decided in advance.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Human coordinates on the floor grid.
5 min	**Watch**	Lesson video; pause on the stage map.
8 min	**Build along**	Falling apple with random start and reset.
15 min	**Perform**	Teleport, hop puzzles, rain-apples Code Lab, stage maths.
8 min	**Probe**	Which way, read it, and the explanation task.
4 min	**Wrap-up**	Exit ticket.

Unplugged warm-up

Human coordinates (5 min): tape an x and y axis on the floor (or draw on the board). Call out positions; students point or walk to them.

Teaching the idea

The Scratch stage is a coordinate plane 480 × 360 with the origin in the centre. “go to x: y:” places a sprite; “change x/y by” moves relatively. Negative y is down. “pick random a to b” gives a random integer. A falling object = forever { change y by −n; if y < bottom then go to random x at top }.

Analogy: Coordinates are like a seat in a cinema: a row (y) and a seat number (x).

Live demo: Show the grid backdrop; drag a sprite and read its x and y aloud. Play “guess where I land” with go to blocks. Then build the falling apple, first without the reset (it gets stuck), then with it.

Questions to ask

Ask	Listen for
Where is (0, 0)?	The middle of the stage.
How do we make something go down?	Change y by a negative number.
Why use pick random?	So the apple appears in a different place each time — less predictable.

Differentiation

Support
Use the grid backdrop and the teleport puzzles first. Give the falling-apple script with the if already built; students add pick random.

Stretch
Make the apple speed up over time (change y by a variable), or add a second apple that falls at a different speed.

Common misconceptions

Misconception	How to fix it
y gets bigger going down	In Scratch, y gets bigger going UP; down is negative.
pick random always gives the same number	It gives a new number each time it runs.
go to x: y: slides the sprite	It jumps instantly; glide slides.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

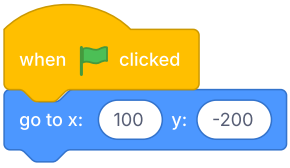
Exit ticket
Write the coordinates of the middle of the stage and the block that moves a sprite down.

Homework
Draw the stage as a grid and mark (0,0), (200,100), (−100,−150).

Answer key — Kitabash activities

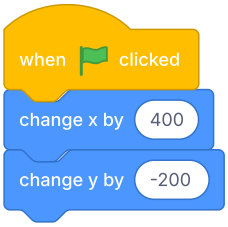
1. Teleport! (perform · block code)

One solution:



2. Hop with x and y (perform · block code)

One solution:

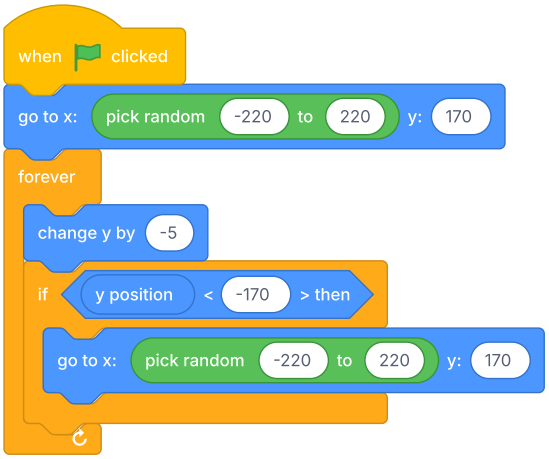


3. Code Lab: make it rain apples (perform · code lab)

Goals checked: Falls down · Starts again at the top · Uses pick random · Uses an if

One solution:

Apple



4. Stage maths (perform · fill blank)

Answers: 480 · 360 · 0

5. Which way? (probe · multiple choice)

Answer: change y by (-10)

6. Read it (probe · multiple choice)

Answer: Moves to a random spot along the TOP of the stage

7. Explain your thinking (probe · free response)

Look for: top, bottom, again. **Example:** If it didn't go back to the top, it would get stuck at the bottom and the game would only have one apple. Going back to the top makes it fall again and again.

Answer key — worksheet

1. **Plot the points:** A: one right, one up. B: two left, middle row. C: middle column, two down.
2. **Up, down, left, right?:** up · left · down
3. **Random:** 1, 3 and 6.
4. **Explain:** So it falls from a different place each time and the game is less predictable.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.